



编译原理（陈火旺第三版）练习答案

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第二章

P-36-6

- (1) L (G) 是 0~9 组成的数字串;
(2) 最左推导:

$N \Rightarrow ND \Rightarrow NDD \Rightarrow NDDD \Rightarrow DDDD \Rightarrow 0DDD \Rightarrow 01DD \Rightarrow 012D \Rightarrow 0127$

$N \Rightarrow ND \Rightarrow DD \Rightarrow 3D \Rightarrow 34$

$N \Rightarrow ND \Rightarrow NDD \Rightarrow DDD \Rightarrow 5DD \Rightarrow 56D \Rightarrow 568$

最右推导:

$N \Rightarrow ND \Rightarrow N7 \Rightarrow ND7 \Rightarrow N27 \Rightarrow ND27 \Rightarrow N127 \Rightarrow D127 \Rightarrow 0127$

$N \Rightarrow ND \Rightarrow N4 \Rightarrow D4 \Rightarrow 34$

$N \Rightarrow ND \Rightarrow N8 \Rightarrow ND8 \Rightarrow N68 \Rightarrow D68 \Rightarrow 568$

P-36-7

G (S): (没有考虑正负符号问题)

$S \rightarrow P|AP$

$P \rightarrow 1|3|5|7|9$

$A \rightarrow AD|N$

$N \rightarrow 2|4|6|8|P$

$D \rightarrow 0|N$

或者: (1) $S \rightarrow A B C | C$

$A \rightarrow 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9$

$B \rightarrow B A | B 0 | \epsilon$

$C \rightarrow 1 | 3 | 5 | 7 | 9$

P-36-8

G (E): $E \rightarrow T|E+T|E-T$

$T \rightarrow F|T*F|T/F$

$F \rightarrow (E) | i$

最左推导:

$E \Rightarrow E+T \Rightarrow T+T \Rightarrow F+T \Rightarrow i+T \Rightarrow i+T*F \Rightarrow i+F*F \Rightarrow i+i*F \Rightarrow i+i*i$

$E \Rightarrow T \Rightarrow T*F \Rightarrow F*F \Rightarrow i*F \Rightarrow i*(E) \Rightarrow i*(E+T) \Rightarrow i*(T+T) \Rightarrow i*(F+T) \Rightarrow i*(i+T) \Rightarrow i*(i+F) \Rightarrow i*(i+i)$

最右推导:

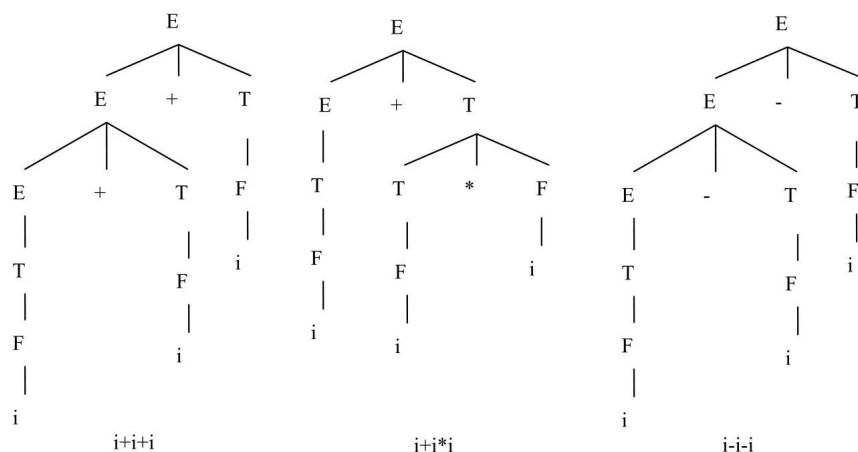
$E \Rightarrow E+T \Rightarrow E+T*F \Rightarrow E+T*i \Rightarrow E+F*i \Rightarrow E+i*i \Rightarrow T+i*i \Rightarrow F+i*i \Rightarrow i+i*i$

$E \Rightarrow T \Rightarrow T*F \Rightarrow T*(E) \Rightarrow T*(E+T) \Rightarrow T*(E+F) \Rightarrow T*(E+i) \Rightarrow T*(T+i) \Rightarrow T*(F+i) \Rightarrow T*(i+i)$

$\Rightarrow F*(i+i) \Rightarrow i*(i+i)$



语法树:



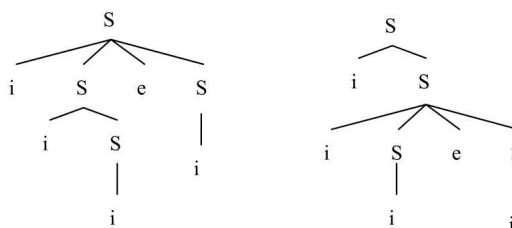
P-36-9

句子: $iiiiei$ 有两个语法树:

$S \Rightarrow iSeS \Rightarrow iSei \Rightarrow iiSei \Rightarrow iiiiei$

$S \Rightarrow iS \Rightarrow iiSeS \Rightarrow iiSei \Rightarrow iiiiei$

因此 $iiiiei$ 是二义性句子, 因此该文法是二义性的。



P-36-10

$S \rightarrow TS | T$

$T \rightarrow (S) | ()$

P-36-11

L1: $G(S): S \rightarrow AC$

$A \rightarrow aAb | ab$

$C \rightarrow cC | \epsilon$

L2: $G(S): S \rightarrow AB$

$A \rightarrow aA | \epsilon$

$B \rightarrow bBc | bc$

L3: $G(S): S \rightarrow AB$

$A \rightarrow aAb | \epsilon$

$B \rightarrow aAb | \epsilon$

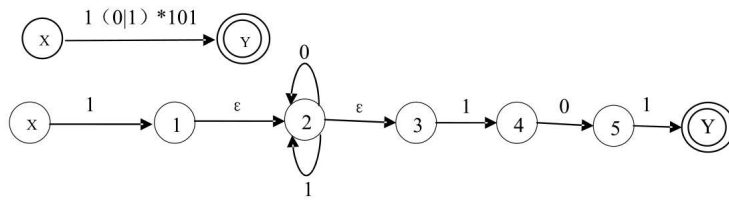
L4: $G(S): S \rightarrow 1S0 | A$

$A \rightarrow 0A1 | \epsilon$

或者: $S \rightarrow A | B \quad A \rightarrow 0A1 | \epsilon \quad B \rightarrow 1B0 | A$

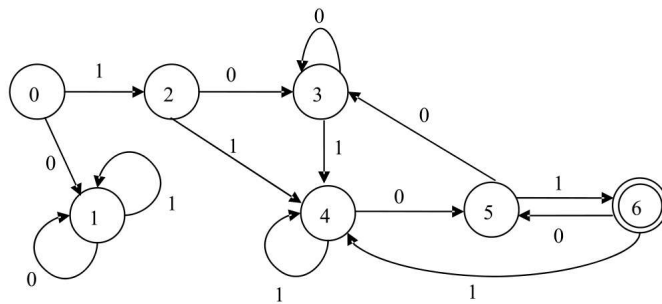
第三章

(1)



确定化:

	0	1
{X}	Φ	{1, 2, 3}
Φ	Φ	Φ
{1, 2, 3}	{2, 3}	{2, 3, 4}
{2, 3}	{2, 3}	{2, 3, 4}
{2, 3, 4}	{2, 3, 5}	{2, 3, 4}
{2, 3, 5}	{2, 3}	{2, 3, 4, Y}
{2, 3, 4, Y}	{2, 3, 5}	{2, 3, 4}



最小化: {0, 1, 2, 3, 4, 5}, {6}

{0, 1, 2, 3, 4, 5}₀={1, 3, 5} {0, 1, 2, 3, 4, 5}₁={1, 2, 4, 6}

{0, 1, 2, 3, 4}, {5}, {6}

{0, 1, 2, 3, 4}₀={1, 3, 5}

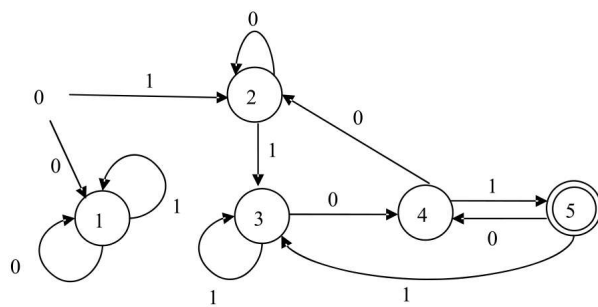
{0, 1, 2, 3}, {4}, {5}, {6}

{0, 1, 2, 3}₀={1, 3} {0, 1, 2, 3}₁={1, 2, 4}

{0, 1}, {2, 3}, {4}, {5}, {6}

{0, 1}₀={1} {0, 1}₁={1, 2} {2, 3}₀={3} {2, 3}₁={4}

{0}, {1}, {2, 3}, {4}, {5}, {6}



P64-8

(1)

$(0|1)^*01$

(2)

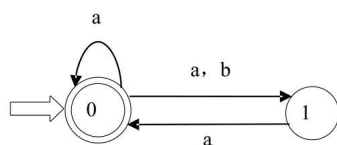
$(1|2|3|4|5|6|7|8|9)(0|1|2|3|4|5|6|7|8|9)^*(0|5) \mid (0|5)$

(3)

$0^*1(0|10^*1)^* \mid 1^*0(1|01^*0)^*$

P84-12

(a)

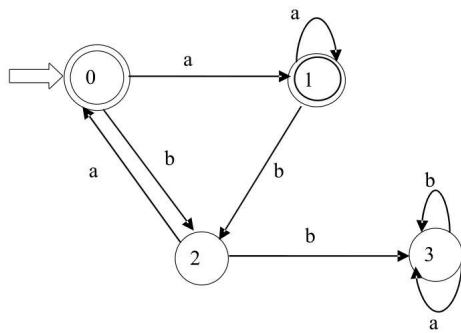


确定化:

	a	b
{0}	{0,1}	{1}
{0,1}	{0,1}	{1}
{1}	{0}	Φ
Φ	Φ	Φ

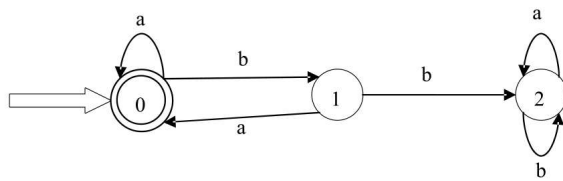
给状态编号:

	a	B
0	1	2
1	1	2
2	0	3
3	3	3



最小化:

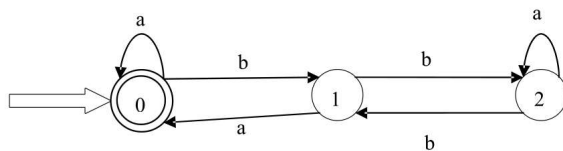
$\{0, 1\}$ $\{2, 3\}$
 $\{0, 1\}_a = \{1\}$, $\{0, 1\}_b = \{2\}$
 $\{2, 3\}_a = \{0, 3\}$, $\{2, 3\}_b = \{3\}$
 $\{0, 1\}$, $\{2\}$, $\{3\}$



(b)

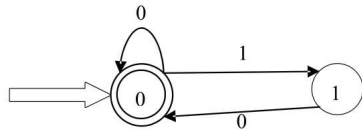
已经确定化, 只需最小化:

$\{0, 1\}$, $\{2, 3, 4, 5\}$
 $\{0, 1\}_a = \{1\}$ $\{0, 1\}_b = \{2, 4\}$
 $\{2, 3, 4, 5\}_a = \{1, 3, 0, 5\}$ $\{2, 3, 4, 5\}_b = \{2, 3, 4, 5\}$
 又: $\{2, 4\}_a = \{1, 0\}$ $\{2, 4\}_b = \{3, 5\}$ $\{3, 5\}_a = \{3, 5\}$ $\{3, 5\}_b = \{2, 4\}$
 分划为: $\{0, 1\}$, $\{2, 4\}$, $\{3, 5\}$
 $\{0, 1\}_a = \{1\}$ $\{0, 1\}_b = \{2, 4\}$
 $\{2, 4\}_a = \{1, 0\}$ $\{2, 4\}_b = \{3, 5\}$
 $\{3, 5\}_a = \{3, 5\}$ $\{3, 5\}_b = \{2, 4\}$
 所以不能再分

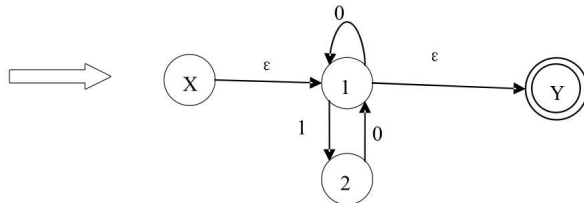


P64-14

正规式: $(0|10)^*$



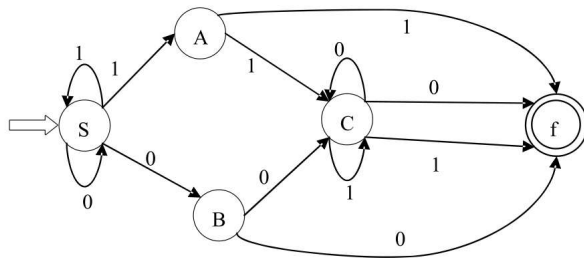
还可以:



然后再确定化, 最小化, 结果应该一样。

P65-15

首先构造 NFA:



则有: $G(f)$

$$f \rightarrow A1 | B0 | C1 | C0$$

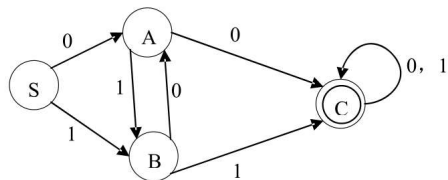
$$C \rightarrow C0 | C1 | A1 | B0$$

$$A \rightarrow S1 | 1$$

$$B \rightarrow S0 | 0$$

$$S \rightarrow S0 | S1 | 0 | 1$$

或者是确定化, 然后最小化:



$G(C)$

$$C \rightarrow C0 | C1 | A0 | B1$$

$$A \rightarrow 0 | B0$$

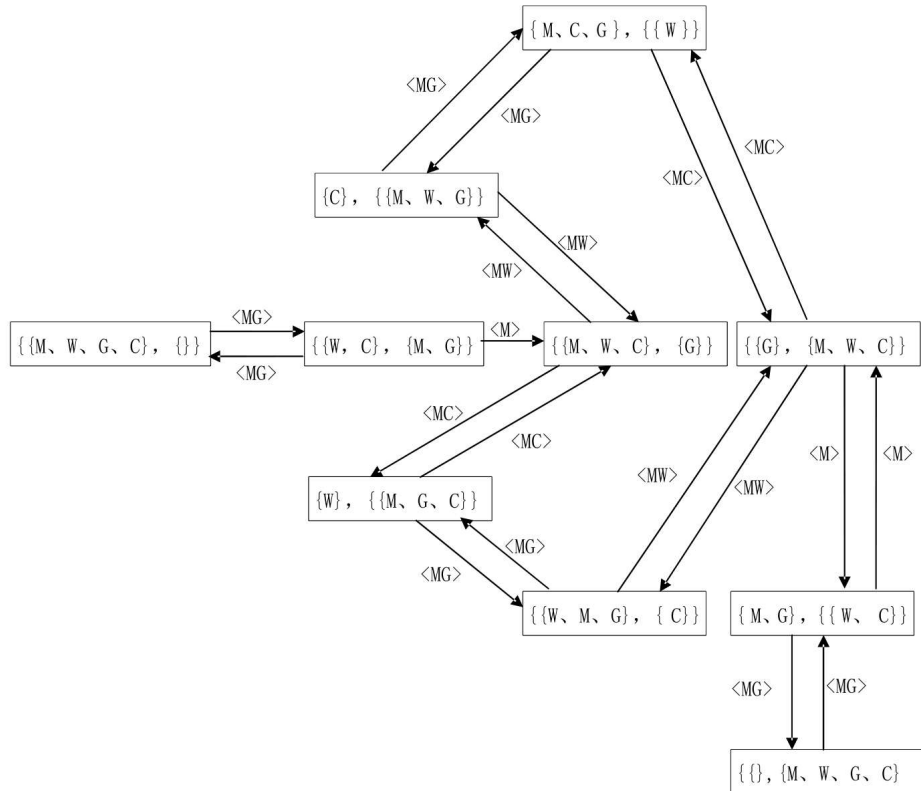
$$B \rightarrow 1 | A1$$

人、狼、羊、白菜：

$\{\{M, W, G, C\}, \{\}\}$ 表示在左岸, $\{\{\}, \{M, W, G, C\}\}$ 在右岸, 将可能存在的状态中去掉不安全状态, 剩下:

$\{\{M, W, G, C\}, \{\}\}$, $\{\{\}, \{M, W, G, C\}\}$, $\{\{M, W, G\}, \{C\}\}$, $\{\{M, W, C\}, \{G\}\}$,
 $\{\{M, G, C\}, \{W\}\}$, $\{\{C\}, \{M, W, G\}\}$, $\{\{G\}, \{M, W, C\}\}$, $\{\{W\}, \{M, G, C\}\}$,
 $\{\{M, G\}, \{W, C\}\}$, $\{\{W, C\}, \{M, G\}\}$

箭弧上的标记符: $\langle M \rangle$: 表示人单独过河、 $\langle MG \rangle$: 表示人和羊过河、 $\langle MW \rangle$: 表示人和狼过河、 $\langle MC \rangle$: 表示人和白菜过河



第四章

P81-1

(1) 按照 T, S 的顺序消除左递归

$G'(S): S \rightarrow a | \wedge | (T)$

$T \rightarrow ST'$

$T' \rightarrow , ST' | \epsilon$

递归下降子程序:

procedure S:

begin

if sym = 'a' or sym= '∧'

then advance

else if sym= '('

then begin

advance; T;

if sym = ')' then advance;

else error;

end

else error

end

procedure T;

begin

S;T'

End

Procedure T' ;

Begin

If sym = ','

Then begin

Advance;

S;T'

End

End

其中: sym 为输入串指针所指的符号; advance 是把输入指针调至下一输入符号。

(2) 求 First 和 Follow 集合:

$First(S) = \{a, \wedge, (\}$ $First(T) = \{a, \wedge, (\}$ $First(T') = \{, , \epsilon\}$

$Follow(S) = \{, ,) , \#\}$ $Follow(T) = \{) \}$ $Follow(T') = \{) \}$

	a	∧	(	)	,	#
S	$S \rightarrow a$	$S \rightarrow \wedge$	$S \rightarrow (T)$			
T	$T \rightarrow ST'$	$T \rightarrow ST'$	$T \rightarrow ST'$			
T'				$T' \rightarrow \epsilon$	$T' \rightarrow , ST'$	



P81-2

文法: $E \rightarrow TE' \quad E' \rightarrow +E \mid \varepsilon \quad T \rightarrow FT' \quad T' \rightarrow T \mid \varepsilon \quad F \rightarrow PF' \quad F' \rightarrow *F' \mid \varepsilon \quad P \rightarrow (E) \mid a \mid b \mid \wedge$

(1)

$\text{First}(E) = \{ (, a, b, \wedge \}$ $\text{First}(E') = \{ +, \varepsilon \}$ $\text{First}(T) = \{ (, a, b, \wedge \}$

$\text{First}(T') = \{ (, a, b, \wedge, \varepsilon \}$ $\text{First}(F) = \{ (, a, b, \wedge \}$ $\text{First}(F') = \{ *, \varepsilon \}$

$\text{First}(P) = \{ (, a, b, \wedge \}$

$\text{Follow}(E) = \{ \#,) \}$ $\text{Follow}(E') = \{ \#,) \}$ $\text{Follow}(T) = \{ +,), \# \}$

$\text{Follow}(T') = \{ +,), \# \}$ $\text{Follow}(F) = \{ +, (, a, b, \wedge,), \# \}$ $\text{Follow}(F') = \{ +, (, a, b, \wedge,), \# \}$

$\text{Follow}(P) = \{ *, +, (, a, b, \wedge,), \# \}$

(2) 文法无左递归, 考察 $E' \rightarrow +E \mid \varepsilon \quad T' \rightarrow T \mid \varepsilon \quad F' \rightarrow *F' \mid \varepsilon \quad P \rightarrow (E) \mid a \mid b \mid \wedge$

$E' \rightarrow +E \mid \varepsilon$: $\text{First}(E') = \{ +, \varepsilon \} \cap \text{Follow}(E') = \{ \#,) \} = \Phi$

$T' \rightarrow T \mid \varepsilon$: $\text{First}(T') = \{ (, a, b, \wedge, \varepsilon \} \cap \text{Follow}(T') = \{ +,), \# \} = \Phi$

$F' \rightarrow *F' \mid \varepsilon$: $\text{First}(F') = \{ *, \varepsilon \} \cap \text{Follow}(F') = \{ (, a, b, \wedge,), \# \} = \Phi$

$P \rightarrow (E) \mid a \mid b \mid \wedge$: 候选式终结首符集两两不相交

所以该文法为 LL(1) 文法。

(3) LL(1) 分析表

	+	*	(	)	a	b	$\wedge$	#
E			$E \rightarrow TE'$		$E \rightarrow TE'$	$E \rightarrow TE'$	$E \rightarrow TE'$	
E'	$E' \rightarrow +E$			$E' \rightarrow \varepsilon$				$E' \rightarrow \varepsilon$
T			$T \rightarrow FT'$		$T \rightarrow FT'$	$T \rightarrow FT'$	$T \rightarrow FT'$	
T'	$T' \rightarrow \varepsilon$		$T' \rightarrow T$	$T' \rightarrow \varepsilon$	$T' \rightarrow T$	$T' \rightarrow T$	$T' \rightarrow T$	$T' \rightarrow \varepsilon$
F			$F \rightarrow PF'$		$F \rightarrow PF'$	$F \rightarrow PF'$	$F \rightarrow PF'$	
F'	$F' \rightarrow \varepsilon$	$F' \rightarrow *F'$	$F' \rightarrow \varepsilon$	$F' \rightarrow \varepsilon$	$F' \rightarrow \varepsilon$	$F' \rightarrow \varepsilon$	$F' \rightarrow \varepsilon$	$F' \rightarrow \varepsilon$
P			$P \rightarrow (E)$		$P \rightarrow a$	$P \rightarrow b$	$P \rightarrow \wedge$	

(4) 构造递归下降程序

Procedure E;

Begin

 If sym = '(' or sym = 'a' or sym = 'b' or sym = ' $\wedge$ '

 Then begin T; E' end

 Else error

End

Procedure E' ;

Begin

 If sym = '+'

 Then begin advance ; E end

 Else if sym <> ')' and sym <> '#' then error

End

Procedure T;

Begin

 If sym = '(' or sym = 'a' or sym = 'b' or sym = ' $\wedge$ '

 Then begin F; T' end

 Else error

End

```

Procedure T';
Begin if sym = '(' or sym = 'a' or sym = 'b' or sym = '^'
    Then begin T;
    Else if sym = '*' then error
End

```

```

Procedure F;
Begin
    if sym = '(' or sym = 'a' or sym = 'b' or sym = '^'
    Then begin P;F' end
    Else error
End

```

```

Procedure F'
Begin
    If sym = '*'
    Then begin advance ; F' end
End

```

```

Procedure P;
Begin
    If sym = 'a' or sym = 'b' or sym = '^'
    Then advance
    Else if sym = '(' then
        Begin advance; E ;
        If sym = ')' then advance
        Else error
    End
    Else error
end

```

P81-3

解答: (1) 该文法不含左递归, 计算 First 集合和 Follow 集合

$\text{Fisrt}(S) = \{a, b, c\}$ $\text{First}(A) = \{a, \varepsilon\}$ $\text{First}(B) = \{b, \varepsilon\}$
 $\text{Follow}(S) = \{\#\}$ $\text{Follow}(A) = \{b, c\}$ $\text{Follow}(B) = \{c\}$
 满足 LL(1) 文法的 3 个条件, 所以是 LL(1) 文法;

(2) 该文法不含左递归, 计算 First 集合和 Follow 集合

$\text{Fisrt}(S) = \{a, b\}$ $\text{First}(A) = \{a, b, \varepsilon\}$ $\text{First}(B) = \{b, \varepsilon\}$
 $\text{Follow}(S) = \{\#\}$ $\text{Follow}(A) = \{b\}$ $\text{Follow}(B) = \{b\}$

考虑 $A \rightarrow a|B|\varepsilon$, $\text{Fisrt}(A)$ 中含有 ε , 而 $\text{Fisrt}(A) \cap \text{Follow}(A) = \{b\}$, 所以不是 LL(1) 文法;

(3) 该文法不含左递归, 计算 First 集合和 Follow 集合

$\text{Fisrt}(S) = \{a, b, \varepsilon\}$ $\text{First}(A) = \{a, \varepsilon\}$ $\text{First}(B) = \{b, \varepsilon\}$
 $\text{Follow}(S) = \{\#\}$ $\text{Follow}(A) = \{a, b, \#\}$ $\text{Follow}(B) = \{a, b, \#\}$

考虑 $A \rightarrow a|\varepsilon$, $\text{Fisrt}(A)$ 中含有 ε , 而 $\text{Fisrt}(A) \cap \text{Follow}(A) = \{a\}$, 所以不是 LL(1) 文法;

(4) 是 LL(1) 文法

P82-4

文法: $\text{Expr} \rightarrow -\text{Expr}$

$\text{Expr} \rightarrow (\text{Expr}) \mid \text{Var ExprTail}$

$\text{ExprTail} \rightarrow -\text{Expr} \mid \varepsilon$

$\text{Var} \rightarrow \text{id VarTail}$

$\text{VarTail} \rightarrow (\text{Expr}) \mid \varepsilon$

解答: $\text{First}(\text{Expr}) = \{-, (, \text{id}\}$ $\text{First}(\text{Var}) = \{\text{id}\}$
 $\text{First}(\text{ExprTail}) = \{-, \varepsilon\}$ $\text{First}(\text{VarTail}) = \{(, \varepsilon\}$
 $\text{Follow}(\text{Expr}) = \{\#, \})\}$ $\text{Follow}(\text{Var}) = \{-, \#, \})\}$
 $\text{Follow}(\text{ExprTail}) = \{\#, \})\}$ $\text{Follow}(\text{VarTail}) = \{-, \#, \})\}$

所以 LL(1) 分析表:

	-	id	(	)	#
Expr	$\text{Expr} \rightarrow -\text{Expr}$	$\text{Expr} \rightarrow \text{Var ExprTail}$	$\text{Expr} \rightarrow (\text{Expr})$		
ExprTail	$\text{ExprTail} \rightarrow -\text{Expr}$			$\text{ExprTail} \rightarrow \varepsilon$	$\text{ExprTail} \rightarrow \varepsilon$
Var		$\text{Var} \rightarrow \text{id VarTail}$			
VarTail	$\text{VarTail} \rightarrow \varepsilon$		$\text{VarTail} \rightarrow (\text{Expr})$	$\text{VarTail} \rightarrow \varepsilon$	$\text{VarTail} \rightarrow \varepsilon$

分析 $\text{id-id}((\text{id}))$

分析栈	输入	所用产生式
#Expr	$\text{id-id}((\text{id})) \#$	
#ExprTail Var	$\text{id-id}((\text{id})) \#$	$\text{Expr} \rightarrow \text{Var ExprTail}$
#ExprTail VarTail id	$\text{id-id}((\text{id})) \#$	$\text{Var} \rightarrow \text{id VarTail}$
#ExprTail VarTail	$-id((\text{id})) \#$	
#ExprTail	$-id((\text{id})) \#$	$\text{VarTail} \rightarrow \varepsilon$
#Expr-	$-id((\text{id})) \#$	$\text{ExprTail} \rightarrow -\text{Expr}$
#Expr	$-id((\text{id})) \#$	
#Expr-	$-id((\text{id})) \#$	$\text{Expr} \rightarrow -\text{Expr}$
#Expr	$\text{id}((\text{id})) \#$	
#ExprTail Var	$\text{id}((\text{id})) \#$	$\text{Expr} \rightarrow \text{Var ExprTail}$
#ExprTail VarTail id	$\text{id}((\text{id})) \#$	$\text{Var} \rightarrow \text{id VarTail}$
#ExprTail VarTail	$((\text{id})) \#$	
#ExprTail)Expr(	$((\text{id})) \#$	$\text{VarTail} \rightarrow (\text{Expr})$
#ExprTail)Expr	$(\text{id})) \#$	
#ExprTail))Expr(	$(\text{id})) \#$	$\text{Expr} \rightarrow (\text{Expr})$
#ExprTail))Expr	$\text{id})) \#$	
#ExprTail))ExprTail Var	$\text{id})) \#$	$\text{Expr} \rightarrow \text{Var ExprTail}$
#ExprTail))ExprTail VarTail id	$\text{id})) \#$	$\text{Var} \rightarrow \text{id VarTail}$
#ExprTail)) ExprTail VarTail	$)) \#$	
#ExprTail)) ExprTail	$)) \#$	$\text{VarTail} \rightarrow \varepsilon$
#ExprTail))	$)) \#$	$\text{ExprTail} \rightarrow \varepsilon$
#ExprTail)	$) \#$	
#ExprTail	$\#$	
#	$\#$	$\text{ExprTail} \rightarrow \varepsilon$

第五章

P133-1

$E \Rightarrow E+T \Rightarrow E+T * F$

短语: $E+T * F$, $T * F$

直接短语: $T * F$

句柄: $T * F$

P133-2

文法: $S \rightarrow a | \wedge | (T)$

$T \rightarrow T, S | S$

(1) 最左推导:

$S \Rightarrow (T) \Rightarrow (T, S) \Rightarrow (S, S) \Rightarrow (a, S) \Rightarrow (a, (T)) \Rightarrow (a, (T, S)) \Rightarrow (a, (S, S)) \Rightarrow (a, (a, S)) \Rightarrow (a, (a, a))$

$S \Rightarrow (T, S) \Rightarrow (S, S) \Rightarrow ((T), S) \Rightarrow ((T, S), S) \Rightarrow ((S, S), S) \Rightarrow (((T), S), S)$

$\Rightarrow (((T, S), S), S) \Rightarrow (((S, S), S), S) \Rightarrow (((a, S), S), S) \Rightarrow (((a, a), S), S) \Rightarrow (((a, a), \wedge, S), S)$

$\Rightarrow (((a, a), \wedge, (T)), S) \Rightarrow (((a, a), \wedge, (S)), S) \Rightarrow (((a, a), \wedge, (a)), S) \Rightarrow (((a, a), \wedge, (T)), a)$

最右推导: $S \Rightarrow (T) \Rightarrow (T, S) \Rightarrow (T, (T)) \Rightarrow (T, (T, S)) \Rightarrow (T, (T, a)) \Rightarrow (T, (S, a))$

$\Rightarrow (T, (a, a)) \Rightarrow (S, (a, a)) \Rightarrow (a, (a, a))$

$S \Rightarrow (T, S) \Rightarrow (T, a) \Rightarrow (S, a) \Rightarrow ((T), a) \Rightarrow ((T, S), a) \Rightarrow ((T, (T)), a) \Rightarrow ((T, (S)), a)$

$\Rightarrow ((T, (a)), a) \Rightarrow ((T, S, (a)), a) \Rightarrow ((T, \wedge, (a)), a) \Rightarrow ((S, \wedge, (a)), a) \Rightarrow ((T), \wedge, (a)), a)$

$\Rightarrow (((T, S), \wedge, (a)), a) \Rightarrow (((T, a), \wedge, (a)), a) \Rightarrow (((S, a), \wedge, (a)), a) \Rightarrow (((a, a), \wedge, (a)), a)$

(2)

$((\underline{a}, a), \wedge, (a)), a)$

$((\underline{S}, a), \wedge, (a)), a)$

$((\underline{T}, a), \wedge, (a)), a)$

$((\underline{T, S}), \wedge, (a)), a)$

$((\underline{(T)}, \wedge, (a)), a)$

$((\underline{S}, \wedge, (a)), a)$

$((T, \wedge, (a)), a)$

$((T, S, (a)), a)$

$((T, (\underline{a})), a)$

$((T, (\underline{S})), a)$

$((T, (\underline{(T)})), a)$

$((T, S), a)$

$(\underline{(T)}, a)$



$(\underline{S}, a)$
 $(T, \underline{a})$
 (T, S)
 (T)
 S

移进归约过程:

步骤	栈	输入串	动作
0	#	$((a, a), \wedge, (a)), a) \#$	初始
1	# (	$((a, a), \wedge, (a)), a) \#$	移进
2	# ((	$(a, a), \wedge, (a)), a) \#$	移进
3	# (((	$\wedge, a), \wedge, (a)), a) \#$	移进
4	# ((a	$, a), \wedge, (a)), a) \#$	移进
5	# (((S	$, a), \wedge, (a)), a) \#$	归约
6	# (((T	$, a), \wedge, (a)), a) \#$	归约
7	# (((T,	$\wedge), \wedge, (a)), a) \#$	移进
8	# (((T, a	$), \wedge, (a)), a) \#$	移进
9	# (((T, S	$), \wedge, (a)), a) \#$	归约
10	# (((T	$), \wedge, (a)), a) \#$	归约
11	# (((T)	$, \wedge, (a)), a) \#$	移进
12	# ((S	$, \wedge, (a)), a) \#$	归约
13	# ((T	$, \wedge, (a)), a) \#$	归约
14	# ((T,	$\wedge, (a)), a) \#$	移进
15	# ((T, \wedge	$, (a)), a) \#$	移进
16	# ((T, S	$, (a)), a) \#$	归约
17	# ((T	$, (a)), a) \#$	归约
18	# ((T,	$(a)), a) \#$	移进
19	# ((T, (	$a)), a) \#$	移进
20	# ((T, (a	$)), a) \#$	移进
21	# ((T, (S	$)), a) \#$	归约
22	# ((T, (T	$)), a) \#$	归约
23	# ((T, (T)	$), a) \#$	移进
24	# ((T, S	$), a) \#$	归约
25	# ((T	$), a) \#$	归约
26	# ((T)	$, a) \#$	移进
27	# (S	$, a) \#$	归约
28	# (T	$, a) \#$	归约
29	# (T,	$a) \#$	移进
30	# (T, a	$) \#$	移进
31	# (T, S	$) \#$	归约
32	# (T	$) \#$	归约
33	# (T)	$\#$	移进
34	#S	$\#$	归约

P133-3: 文法: $G(S): S \rightarrow a \mid \wedge \mid (T) \quad T \rightarrow T, S \mid S$

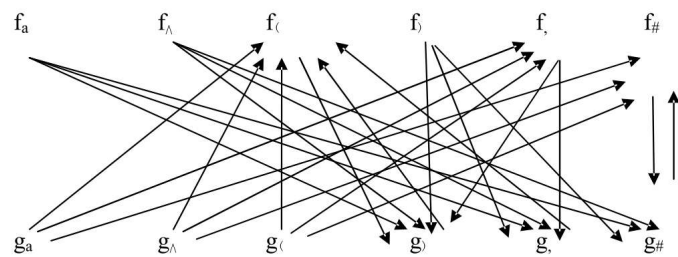
(1) $FIRSTVT(S) = \{ a, \wedge, (\}$ $FIRSTVT(T) = \{ ,, a, \wedge, (\}$
 $LASTVT(S) = \{ a, \wedge,) \}$ $LASTVT(T) = \{ ,, a, \wedge,) \}$

(2) 算符优先分析表

	a	$\wedge$	(	)	,	#
a				$\succ$	$\succ$	$\succ$
$\wedge$				$\succ$	$\succ$	$\succ$
(	$\prec$	$\prec$	$\prec$	=	$\prec$	
)				$\succ$	$\succ$	$\succ$
,	$\prec$	$\prec$	$\prec$	$\succ$	$\succ$	
#	$\prec$	$\prec$	$\prec$			=

(3) 优先函数:

	a	$\wedge$	(	)	,	#
f	6	6	2	6	4	2
g	7	7	7	2	3	2



如果不考虑#, 则: 优先函数:

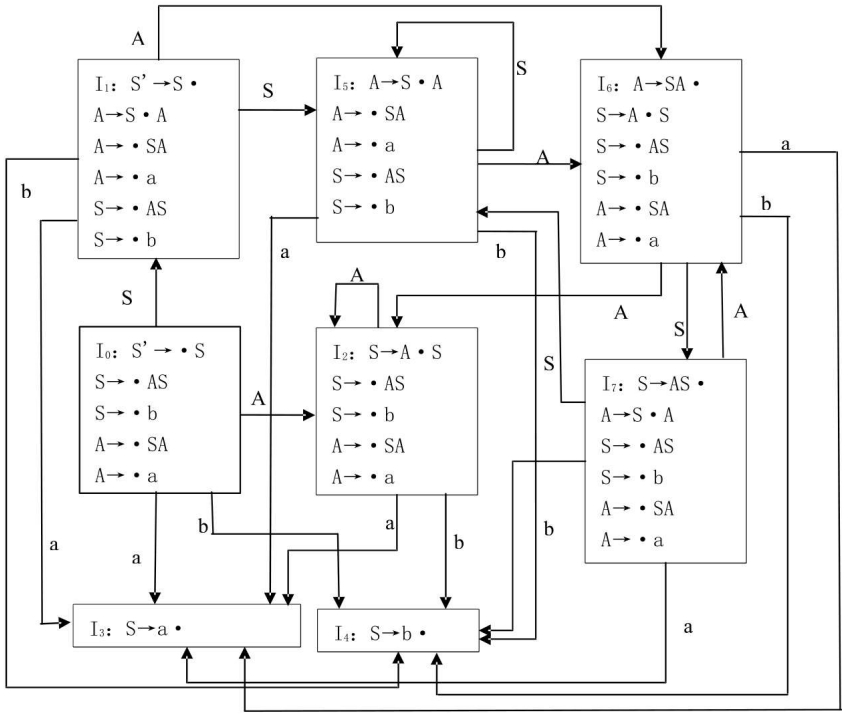
	a	$\wedge$	(	)	,
f	4	4	2	4	4
g	5	5	5	2	3

分析过程:

栈	输入	
#	(a, (a, a)) #	初始
#(	a, (a, a) #	移进
#(a	, (a, a) #	移进
#(S	, (a, a) #	归约
#(S,	(a, a) #	移进
#(S, (	a, a) #	移进
#(S, (a	, a) #	移进
#(S, (S	, a) #	归约
#(S, (S,	a) #	移进

#(S, (S, a	)) #	移进
#(S, (S, S	)) #	归约
#(S, (T	)) #	归约
#(S, (T)	) #	移进
#(S, S	) #	归约
#(T	) #	归约
#(T)	#	移进
#S	#	归约

P134-5
(1)



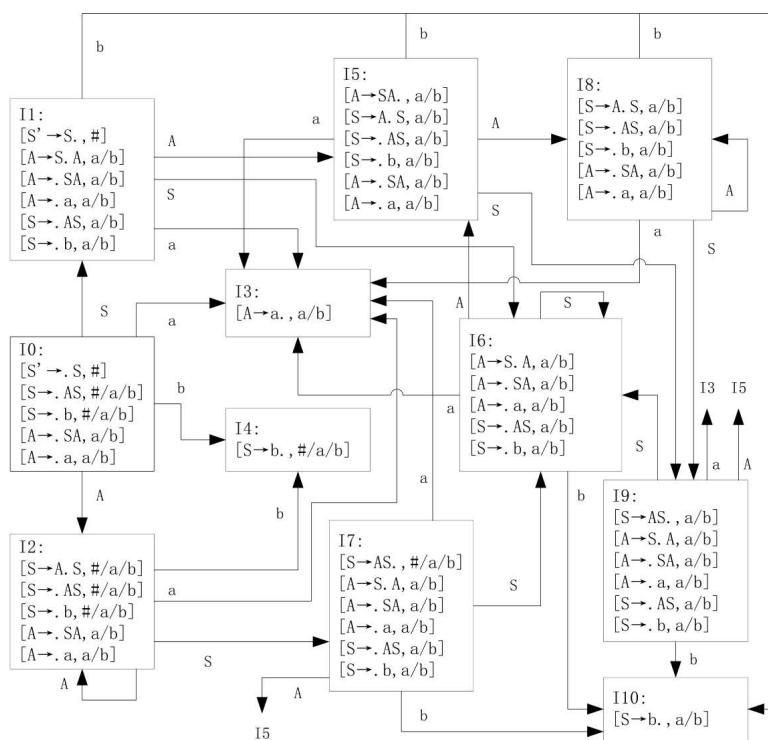
考察 I_1 、 I_6 、 I_7 :

I_1 : 存在移进-归约冲突, 因为 $\text{Follow}(S') = \{\#\}$, 不包含 a 或 b , 因此冲突可以使用 SLR 解决方法解决。

I_6 : 存在移进-归约冲突, 因为 $\text{Follow}(A) = \{a, b\}$, 因此无法使用 SLR 方法解决移进-归约冲突

I_7 : 存在移进-归约冲突, 因为 $\text{Follow}(S) = \{\#, a, b\}$, 因此无法解决移进-归约冲突
所以不是 SLR(1) 文法。

构造 LR(1) 项目集规范族:



检查 I5, $[A \rightarrow SA. , a/b]$, 要求输入为 a 或者 b 使用 $A \rightarrow SA$ 归约, 而 $[S \rightarrow . b, a/b]$ 及 $[A \rightarrow . a, a/b]$ 要求移进, 因此存在移进-归约冲突, 所以不是 LR (1) 文法。

P135-8

解答:

不存在左递归;

因为 $\text{First}(AaAb) = \{a\}$, $\text{First}(BbBa) = \{b\}$ 所以交集为空

所以该文法是 LL (1) 文法。

$I_0 = \{S \rightarrow . AaAb, S \rightarrow . BbBa, A \rightarrow . , B \rightarrow . \}$

$I_1 = G_0(I_0, A) = \{S \rightarrow A. aAb\}$

$I_2 = G_0(I_0, B) = \{S \rightarrow B. bBa\}$

$I_3 = G_0(I_1, a) = \{S \rightarrow Aa. Ab, A \rightarrow . \}$

$I_4 = G_0(I_2, b) = \{S \rightarrow Bb. Ba, B \rightarrow . \}$

$I_5 = G_0(I_3, A) = \{S \rightarrow AaA. b\}$

$I_6 = G_0(I_4, B) = \{S \rightarrow BbB. a\}$

$I_7 = G_0(I_5, b) = \{S \rightarrow AaAb. \}$

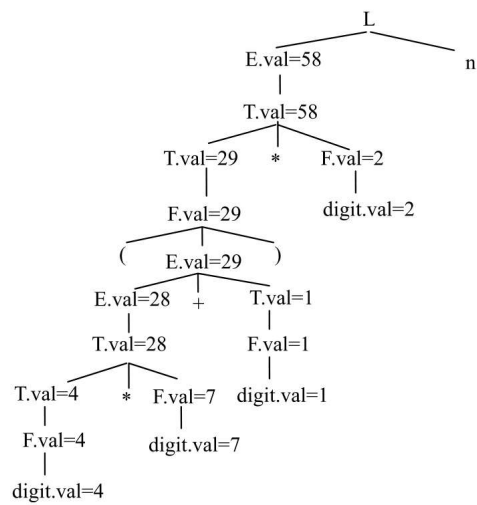
$I_8 = G_0(I_6, a) = \{S \rightarrow BbBa. \}$

考虑: I_0 : 存在两个归约项目, $A \rightarrow . , B \rightarrow .$, $\text{Follow}(A) = \{a, b\}$, $\text{Follow}(B) = \{a, b\}$, 所以冲突不能解决, 不是 SLR (1) 文法。

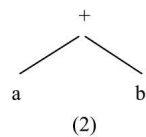
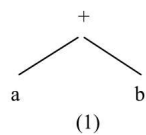
第六章

P164-1

解答：表达式 $(4*7+1)*2$ 的附注语法树：



P164-2



p165-5

(1)

$E \rightarrow E_1 + T$ { if $(E_1.type = int) \text{ and } (T.type = int)$ then $E.type = int$
 Else $E.type = real$ }

$E \rightarrow T$ { $E.type = T.type$ }

$T \rightarrow num. \text{ num}$ { $T.type = real$ }

$T \rightarrow num$ { $T.type = int$ }

(2)

$E \rightarrow E_1 + T$ { if $(E_1.type = int) \text{ and } (T.type = int)$ then
 $E.type = int$
 $E.code = E_1.code \parallel T.code \parallel +$
 Else if $(E_1.type = real) \text{ and } (T.type = real)$
 $E.type = real$
 $E.code = E_1.code \parallel T.code \parallel +$
 Else if $E_1.type = int$ then
 $E.type = real$
 $E.code = E_1.code \parallel intto real \parallel T.code \parallel +$



```

                Else
                    E.type = real
                    E.code = E1.code || T.code || inttoreal || +
                End if
E → T           { E.type = T.type
                  E.code = T.code }
T → num. num    { T.type = real
                  E.code = num.num }
T → num         { T.type = int
                  E.code = num }

```

P164-7

```

S → L1.L2      { S.val = L1.val + L2.val / 2L2.length }
S → L           { S.val = L.val }
L → L1B         { L.val = 2*L1.val + B.c
                  L.length = L1.length + 1 }
L → B           { L.val = B.c
                  L.length = 1 }
B → 0           { B.c = 0 }
B → 1           { B.c = 1 }

```

P165-11

对 D, L, T 设置综合属性 type。过程 addtype (id.entry, type) 用来将标识符 id 的类型 type 填入到符号表中。

(1) 翻译模式:

```

D → id L        { addtype (id.entry, L.type) }
L → , id L1     { L.type = L1.type ; addtype (id.entry, L1.type) }
L → :T          { L.type = T.type }
T → integer     { T.type = integer }
T → real        { T.type = real }

```

(2) 假设 Ttype 为已定义的表示“类型”的数据结构，预测翻译器如下:

```

procedure D;
  var l_type: Ttype
  begin
    if sym = "id" then
      begin
        advance ;
        l_type = L ;
        addtype(id.entry , l_type)
      end
    else error
  end;

```

```

procedure L;
  var l_type:Ttype;
begin
  if sym = “,” then
    begin
      advance;
      if sym = “id” then
        begin
          advance ;
          l_type = L ;
          adddtype(id.entry, l_type)
        end
      else error ;
    end
  else if sym = “.” then
    begin
      advance ;
      l_type = T ;
    end
  else error ;
  return (l_type) ;
end;

procedure T ;
  var t_type: Ttype ;
begin
  if sym = “integer” then
    begin
      advance ;
      t_type = integer ;
    end
  else if sym = “real” then
    begin
      advance ;
      t_type = real ;
    end
  else error
  return(t_type);
end;

```

第七章

P217-1

$a * (-b + c)$ 后缀式: $ab - c + *$

$a + b * (c + d / e)$ 后缀式: $abcde / + * +$

$-a + b * (-c + d)$ 后缀式: $a - bc - d + * +$

not A or not (C or not D) 后缀式: $A \text{ not } C \text{ D not or not or}$

(A and B) or (not C or D) 后缀式: $A \text{ B and } C \text{ not D or or}$

(A or B) and (C or not D and E) 后缀式: $A \text{ B or } C \text{ D not E and or and}$

if $(x + y) * z = 0$ then $(a + b) \uparrow c$ else $a \uparrow b \uparrow c$ 后缀式: $xy + z * 0 = ab + c \uparrow abc \uparrow \uparrow$ if-then-else

P217-3

$-(a + b) * (c + d) - (a + b + c)$

三元式:

(1) +, a, b

(2) -, (1), -

(3) +, c, d

(4) *, (2), (3)

(5) +, a, b

(6) +, (5), c

(7) -, (4), (6)

间接三元式:

三元式表:

(1) +, a, b

(2) -, (1), -

(3) +, c, d

(4) *, (2), (3)

(5) +, (1), c

(6) -, (4), (5)

间接码表: (1), (2), (3), (4), (1), (5), (6)

四元式序列:

(1) +, a, b, T1

(2) -, T1, -, T2

(3) +, c, d, T3

(4) *, T2, T3, T4

(5) +, a, b, T5

(6) +, T5, c, T6

(7) -, T4, T6, T7



P218-8

自下而上分析过程中把赋值语句 $A := B * (-C + D)$ 翻译成四元式的步骤:

步骤	输入串	栈	PLACE	四元式
(1)	$A := B * (-C + D)$			
(2)	$:= B * (-C + D)$	i	A	
(3)	$B * (-C + D)$	i :=	A-	
(4)	$* (-C + D)$	i := i	A-B	
(5)	$* (-C + D)$	i := E	A-B	
(6)	$(-C + D)$	i := E*	A-B-	
(7)	$-C + D)$	i := E* (	A-B--	
(8)	$C + D)$	i := E* (-	A-B---	
(9)	$+ D)$	i := E* (-i	A-B---C	
(10)	$+ D)$	i := E* (-E	A-B---C	(-, C, -, T1)
(11)	$+ D)$	i := E* (E	A-B--T1	
(12)	$D)$	i := E* (E+	A-B--T1-	
(13)	$)$	i := E* (E+i	A-B--T1-D	
(14)	$)$	i := E* (E+E	A-B--T1-D	(+, T1, D, T2)
(15)	$)$	i := E* (E	A-B--T2	
(16)		i := E* (E)	A-B--T2-	
(17)		i := E*E	A-B-T2	(*, B, T2, T3)
(18)		i := E	A-T3	(:=, T3, -, A)
(19)		A		

P218-5

设 A、B 为 10 x 20 的数组，C、D 大小为 10 的数组，数组每维下届为 1，每个数据项宽度为 4，则:

$A[i, j] := B[i, j] + C[A[k, 1]] + D[i+j]$

$T1 := i * 20$

$T1 := T1 + j$

$T2 := A - 84$

$T3 := 4 * T1$

$T4 := i * 20$

$T4 := T4 + j$

$T5 := B - 84$

$T6 := 4 * T4$

$T7 := T5[T6]$

$T8 := k * 20$

$T8 = T8 + 1$

$T9 := A - 84$

$T10 := 4 * T8$

$T11 := T9[T10]$

$T12 := C - 4$

$T13 := 4 * T11$

$T14 := T12[T13]$



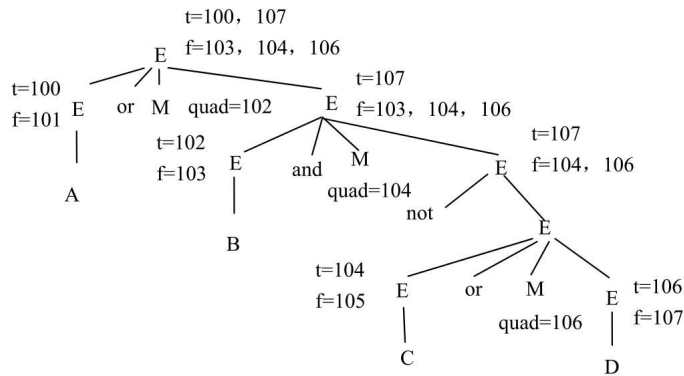
```

T15 := T7 + T14
T16 := i + j
T17 := D - 4
T18 := 4 * T16
T19 := T17[T18]
T20 := T15 + T19
T2[T3] := T20

```

P218-6

A or (B and not (C or D)):

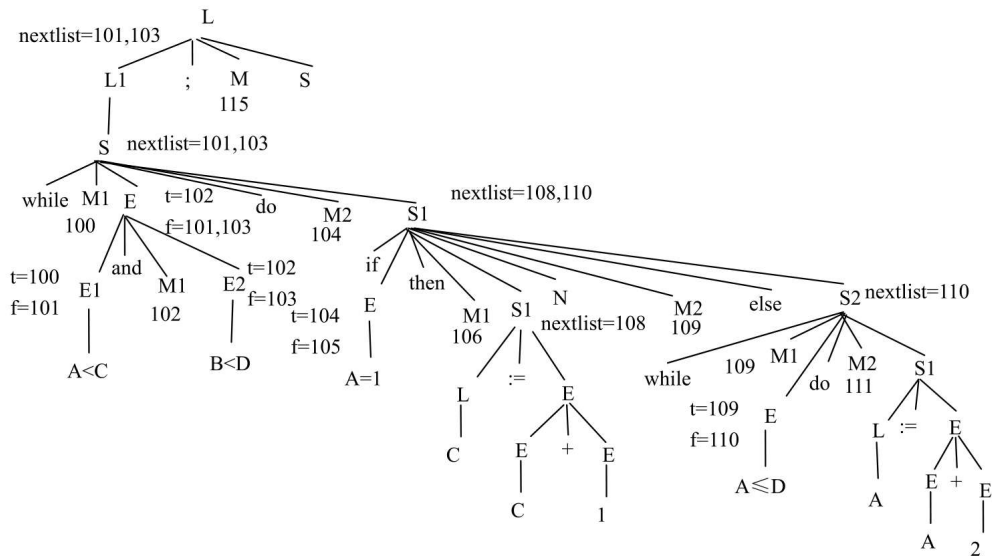


```

100: (jnz, A, -, 0)
101: (j, -, -, 102)
102: (jnz, B, -, 104)
103: (j, -, -, 0)
104: (jnz, C, -, 0)
105: (j, -, -, 106)
106: (jnz, D, -, 0)
107: (j, -, -, 0)

```

P218-7



100: (j<, A, C, 102)
 101: (j, -, -, 115)
 102: (j<, B, D, 104)
 103: (j, -, -, 115)
 104: (j=, A, '1', 106)
 105: (j, -, -, 109)
 106: (+, C, '1', T1)
 107: (:=, T1, -, C)
 108: (j, -, -, 100)
 109: (j≤, A, D, 111)
 110: (j, -, -, 100)
 111: (+, A, '2', T2)
 112: (:=, T2, -, A)
 113: (j, -, -, 109)
 114: (j, -, -, 100)
 115:

P219-12

(1) 如果该程序执行, 则先会打印出:

```
MAXINT-5
MAXINT-4
MAXINT-3
MAXINT-2
MAXINT-1
MAXINT
```

然后对于有些可能出现的整型数溢出而出现运行时的异常。

(2) 根据其语义, 先确定 PASCAL 语言 for 语句的中间代码结构如下:

```
t1 := initial
t2 := final
if t1 > t2 goto L2
v := t1
L1: S 的代码
    if v = t2 goto L2
    v := v + 1
    goto L1
```

L2:

为了便于语法制导的翻译, 将 PASCAL 语言的 for 语句:

$S \rightarrow \text{for } V := E1 \text{ to } E2 \text{ do } S1$

改写成如下产生式:

$S \rightarrow F \text{ do } S1$

$F \rightarrow \text{for } v := E1 \text{ to } E2$

翻译模式如下:

$F \rightarrow \text{for } v := E1 \text{ to } E2$

```
{ F.nextlist := makelist(nextquad);
  emit (j>, E1.place, E2.place, 0);
  emit (:=, E1.place, -, v.place);
  F.quad := nextquad;
  F.place1 := E2.place;
  F.place2 := entry (v); }
```

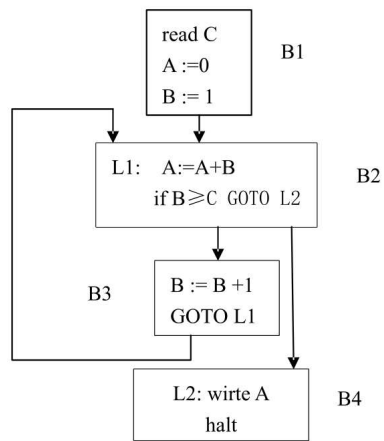
$S \rightarrow F \text{ do } S1$

```
{ backpatch (S1.nextlist, F.quad);
  S.nextlist := merge (F.nextlist, makelist (nextquad));
  emit(j=, F.place1, F.place2, 0);
  emit (+, F.place2, 1, F.place2);
  emit (j, -, -, F.quad) }
```

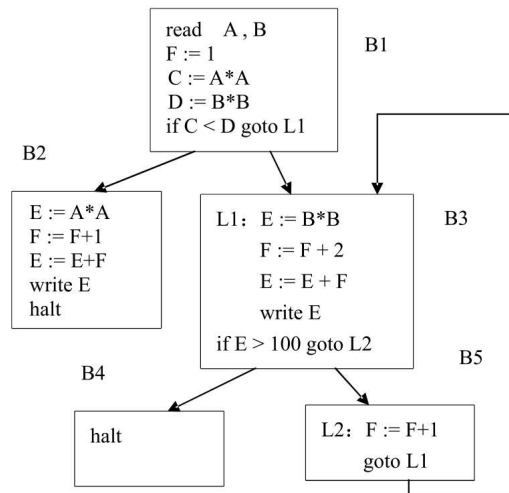

第十章

P306-1:
 read C
 A := 0
 B := 1
 L1: A := A + B
 if B $\geq$ C GOTO L2

 B := B + 1
 GOTO L1
 L2: write A
 halt



P306-2
 read A, B
 F := 1
 C := A * A
 D := B * B
 if C < D goto L1
 E := A * A
 F := F + 1
 E := E + F
 write E
 halt
 L1: E := B * B
 F := F + 2
 E := E + F
 write E
 if E > 100 goto L2
 halt
 L2: F := F + 1
 goto L1



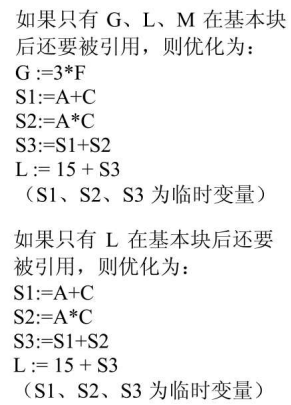
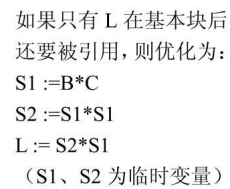
```

B1:  A := B*C
      D := B/C
      E := A+D
      F := 2*E
      G := B*C
      H := G*G
      F := H*G
      L := F
      M := L

```

如果只有 G、L、M 在基本块后还要被引用，则优化为：

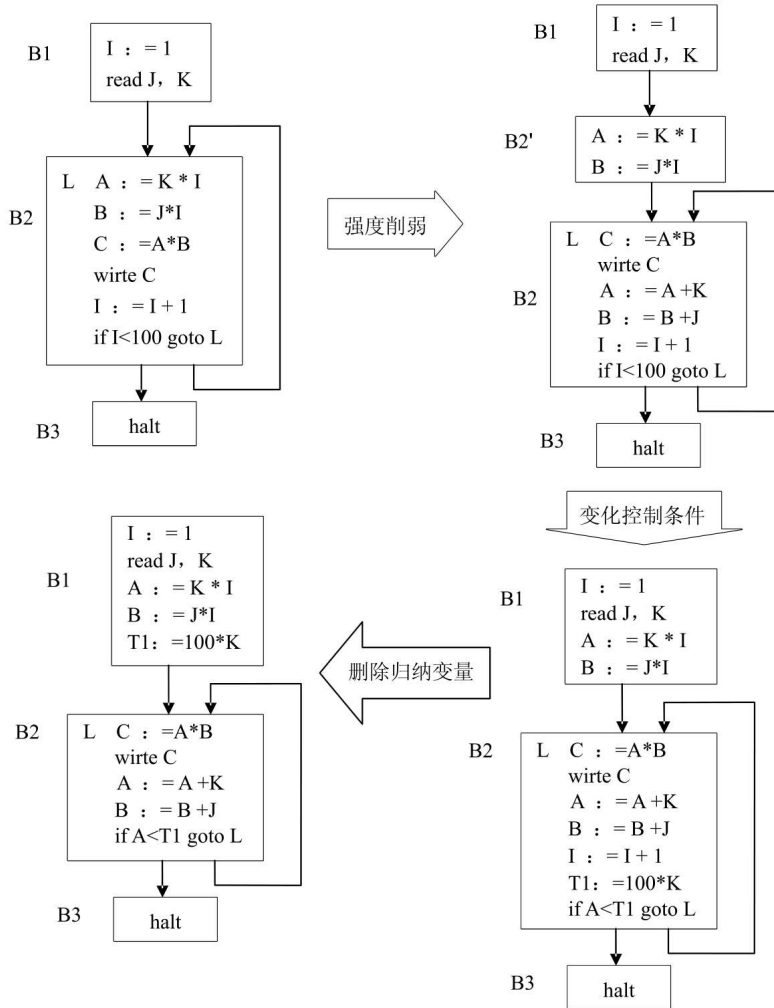
```
G := B * C
S1 := G * G
L := S1 * G
M := L
(S1 为临时变量)
```



P307-4 对一下四元式程序，对其中循环进行循环优化：

```
I := 1
read J, K
L:  A := K * I
    B := J * I
    C := A * B
    write C
    I := I + 1
    if I < 100 goto L
halt
```

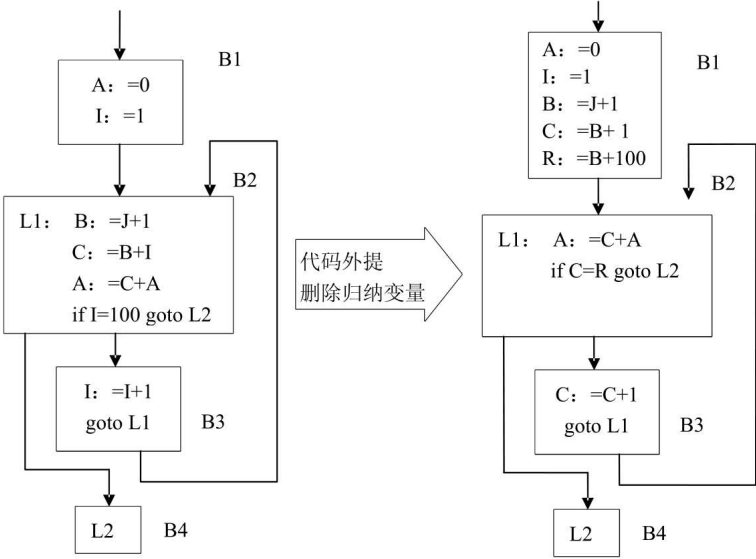
解答：首先进行基本块划分，画出程序流程图：从图中可以看出需要优化的循环块为 B2



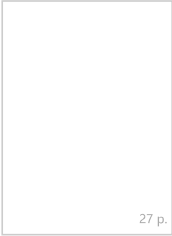
P307-5 以下是某程序的最内循环模式对其进行循环优化。

```
A : = 0
I : = 1
L1:  B: =J+1
      C: =B+I
      A: =C+A
      if I=100 goto L2
      I : =I+1
      goto  L1
L2:
```

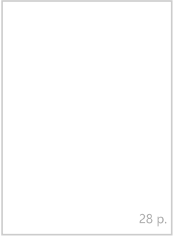
解答：首先做出程序流程图，然后进行优化：



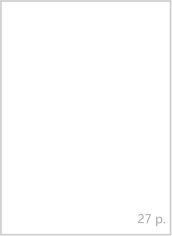
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